Meridian — validation report

Frozen-model deployment after the training period, the Meridian-WM world-model network, and the leak / bluff / fake audit

Generated 2026-11-11 01:50 UTC from commit **b522968** of branch `claude/sleepy-cray-ti3mxs`. Audit: **22 PASS, 1 WARN, 0 FAIL, 0 SKIP**.

Every number in this document is read from a results file written by the pipeline; the verdicts are computed from those numbers with the bar that was fixed before any result was seen (PREREGISTRATION.md §5: QLIKE at least 5% below the reference *and* date-level Diebold–Mariano p < 0.05). The test plan for the overnight runs (Amendment 8) was committed to git before any of them started; audit check X1 verifies the timestamps.

1. Executive summary

Test	Model	Window (target dates)	vs reference	95% CI	DM p	Verdict vs pre-registered bar
Deploy A	Meridian M0 (audited)	2026-11-02 → 2026-11-04 (3)	-8.6%	—	—	Not testable (< 20 dates)
V1/V3	Meridian legacy, level offset	2026-08-03 → 2026-11-04 (68)	-1.5%	[-5.9, +4.7]	0.349	Not significant
V1/V3	Meridian legacy, Jensen term	2026-08-03 → 2026-11-04 (68)	+16.8%	[-3.7, +35.5]	0.954	Significantly WORSE
V1/V3	Meridian-WM (frozen)	2026-08-03 → 2026-11-04 (68)	+7.9%	[-8.0, +21.5]	0.877	Not significant
V2 primary	Meridian-WM as written	walk-forward, fold 7 (test from 2019-04-25)	—	—	—	FAILED: diverged to NaN (3/3 seeds)
V2 (8a)	Meridian-WM (walk-forward)	2019-01-01 → 2026-06-29 (2168)	+31.6%	[+26.1, +38.4]	1.000	Significantly WORSE
V2 (8a)	Meridian M0 (audited)	2019-01-01 → 2026-06-29 (2168)	-6.7%	[-9.7, -4.5]	<0.001	MEETS the bar
V2 (8a)	Meridian-WM, recalibrated	2019-01-01 → 2026-06-29 (2168)	+6.8%	[+3.1, +11.2]	1.000	Significantly WORSE
Live log	Meridian M0 (pre-committed)	2026-10-30 → 2026-11-09 (4 origins)	+3.4%	—	—	Not testable (< 20 dates)

Reference = per-asset HAR-RV fit at the same cut-off (recalibrated HAR for the recalibrated row). Negative % = lower QLIKE = better. 'Window' counts scored target dates (origins for the live log).

What this means.

• After their training ended, no Meridian or Meridian-WM model beat HAR by a statistically established margin. Over the 68 post-training dates of V1/V3 the best Meridian variant (legacy + level offset) is -1.5% (DM p 0.349); the frozen Meridian-WM is +7.9% (p 0.877). The best model in that window is Ridge (-3.0%, p 0.093).

• In the long walk-forward test (V2, 2168 holdout dates, same folds as the audited benchmark) Meridian-WM is +31.6% vs HAR (p 1.000), while the audited Meridian M0 is -6.7% (p <0.001) on exactly the same rows. Verdict for the WM network: Significantly WORSE.

• With FX rows aligned to their true trading day (audit D4) the audited benchmark's headline M0 result becomes -6.7% [-9.6, -4.5], DM p <0.001 (as stored: -6.7% [-9.7, -4.5], p <0.001).

• 'MEETS the bar' for M0 refers to its point estimate (-6.7%); the upper end of its 95% interval ([-9.7, -4.5]) is above -5%, so the size of the gain is uncertain even though the gain itself is clear. That holdout result predates this work and is reproduced exactly here, not newly established.

• The audited M0 has only 3 target dates after its own training cut-off, so its post-training performance is not yet testable; its live pre-committed log covers 4 origins. Months of daily logging are needed before a forward verdict exists.

• The audit found no data leak, no changed weights, no look-ahead, and no number in a report that differs from its source file. It did find, and the work fixed, six real defects, including a pre-existing FX date-labelling issue that overstated some significance levels (Section 10).

2. What was asked and what was done

• **1. What is the WM component?** Answered in Section 3.

• **2. Deploy without additional training, on the period after training ends.** The two frozen Meridian snapshots saved in the repository were loaded (sha256-checked) and only run forward: Snapshot A (audited M0, trained through 2026-10-30) and Snapshot B (pre-audit legacy model, trained through late July 2026). No weight was changed; audit W1/W2 prove it.

• **3. Measure and benchmark against other models.** 17 comparison models (HAR family, GARCH family, EWMA, random walk, Ridge, random forest, LightGBM, TimeMixer) were fit once on exactly the same pre-cut-off data and then also run forward frozen; tail risk (VaR/ES) was benchmarked against GJR-GARCH + FHS.

• **4. Audit for leaks, bluffed numbers and fakes.** A 22-check audit (Section 9) runs on every push in CI.

• **Deeper overnight validation (Amendment 8, pre-registered).** V1: Snapshot B over the longest honest post-training window, with tails. V2: the Meridian-WM network inside the audited walk-forward harness. V3: Meridian-WM trained once at the V1 cut-off, frozen, and deployed like V1.

3. The WM component

Meridian-WM (`meridian/meridian_wm.py`, class `MeridianWM`) is one network with four parts: (1) a sticky switching state-space belief core (3 soft regimes; regime = most likely mode) with its own volatility head; (2) a separate diagonal state-space volatility core whose EMA-smoothed belief feeds the main next-day log-variance head (QLIKE loss); (3) a Student-t head over next-day returns (tail heaviness and scale); (4) a JEPA latent predictor with SIGReg regularization whose prediction error is the 'surprise' score. The wider *world model* (WORLD_MODEL.md) is a bank of separately trained modules (volatility, tail, regime, factor, connectedness, propagation, covariance) linked by inspectable combiners.

No trained Meridian-WM weights existed in the repository (every script retrained it and discarded it), so it could not be deployed without training. To validate it anyway, Amendment 8 pre-registered two tests that train it only on data before each forecast: V2 (walk-forward, retrained per fold exactly like every benchmark model) and V3 (trained once on labels through 2026-07-29, frozen with sha256 hashes, then run forward untouched).

4. Data and provenance

Daily OHLC from Yahoo (the same source the models were trained on) and VIX / 10y / 2y yields from FRED. Each asset was checked against the next-day realized variance stored by the audited benchmark on every overlapping date (audit D1).

Asset	Source	First	Last	Overlap days	Median |Δ log RV|	Correlation
SPY	yahoo	2007-01-03	2026-11-04	3794	0.0000	1.0000
QQQ	yahoo	2007-01-03	2026-11-04	3794	0.0000	1.0000
IWM	yahoo	2007-01-03	2026-11-04	3794	0.0000	1.0000
AAPL	yahoo	2007-01-03	2026-11-04	3794	0.0000	0.9999
MSFT	yahoo	2007-01-03	2026-11-04	3794	0.0000	1.0000
JPM	yahoo	2007-01-03	2026-11-04	3794	0.0000	1.0000
XOM	yahoo	2007-01-03	2026-11-04	3794	0.0000	1.0000
JNJ	yahoo	2007-01-03	2026-11-04	3794	0.0000	1.0000
TLT	yahoo	2007-01-03	2026-11-04	3794	0.0000	0.9999
GLD	yahoo	2007-01-03	2026-11-04	3794	0.0000	1.0000
EURUSD	yahoo	2007-01-01	2026-11-04	3926	0.0000	1.0000
USDJPY	yahoo	2007-01-01	2026-11-04	3926	0.0000	1.0000
GBPUSD	yahoo	2007-01-01	2026-11-04	3925	0.0000	1.0000

Universe: the 11 pre-registered core assets (8 US equities/ETFs + EURUSD, USDJPY, GBPUSD) are scored; TLT and GLD are a separate out-of-universe check. Target: next-day Garman–Klass variance. Loss: QLIKE.

5. First deployment: October and November 2026

Snapshot A — Meridian M0 forward-frozen (audited config, 5 seeds); training labels end 2026-10-30; Nov-2026: 3 target dates.

Rank	Model	QLIKE	vs ref.	95% CI	DM p	MCS 90%	Assets beating ref.
1	Meridian M0, real-time VIX	0.2763	-11.5%	—	—	—	9/11
2	Meridian M0 (audited)	0.2853	-8.6%	—	—	—	8/11
3	Random forest	0.2868	-8.2%	—	—	—	9/11
4	Ridge	0.2908	-6.9%	—	—	—	8/11
5	LHAR	0.2953	-5.4%	—	—	—	8/11
6	LightGBM	0.3083	-1.3%	—	—	—	7/11
9	HAR (frozen with M0)	0.3123	-0.0%	—	—	—	6/11
10	HAR-RV per asset (ref.)	0.3123	+0.0%	—	—	—	0/11
14	EWMA	0.3577	+14.5%	—	—	—	3/11
16	GJR-GARCH	0.4329	+38.6%	—	—	—	4/11

Snapshot B — Meridian legacy final (pre-audit config, 1 seed); training labels end 2026-09-08; Oct-2026: 25 target dates.

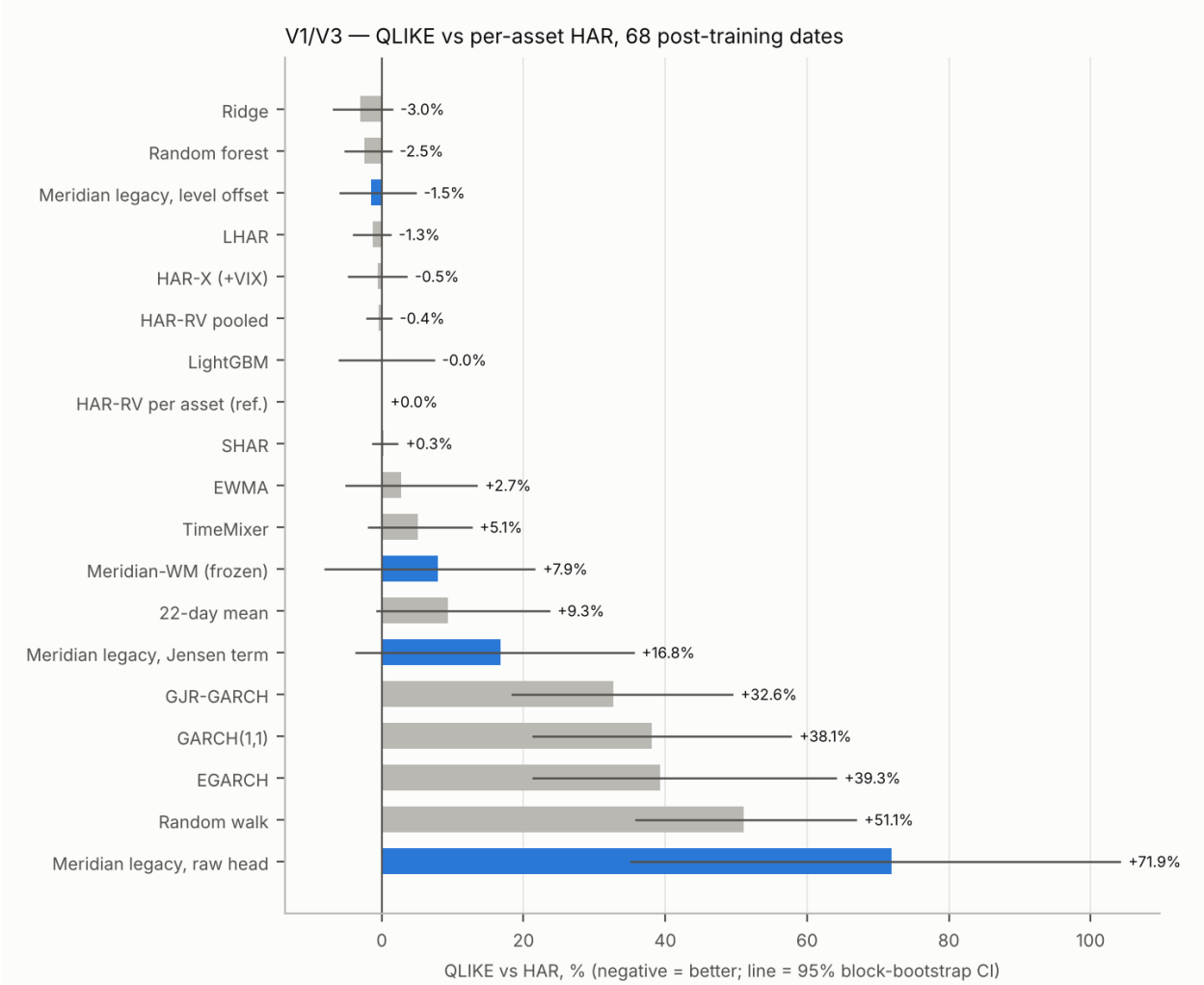
Rank	Model	QLIKE	vs ref.	95% CI	DM p	MCS 90%	Assets beating ref.
2	EWMA	0.3061	-5.3%	[-13.2, +4.9]	0.206	—	8/11
3	Random forest	0.3138	-2.9%	[-7.3, +0.7]	0.114	—	8/11
4	Ridge	0.3144	-2.7%	[-6.1, +1.3]	0.013	—	8/11
7	LHAR	0.3221	-0.3%	[-2.9, +1.9]	0.435	—	7/11
8	LightGBM	0.3224	-0.2%	[-5.6, +7.1]	0.358	—	7/11
9	HAR-RV per asset (ref.)	0.3232	+0.0%	—	—	—	0/11
11	Meridian legacy, level offset	0.3332	+3.1%	[-2.7, +7.7]	0.841	—	5/11
13	GJR-GARCH	0.3464	+7.2%	[-3.2, +22.0]	0.982	—	7/11
16	Meridian legacy, Jensen term	0.4292	+32.8%	[-1.0, +47.0]	0.941	—	2/11

Rank	Model	QLIKE	vs ref.	95% CI	DM p	MCS 90%	Assets beating ref.
18	Meridian legacy, raw head	0.6384	+97.5%	[+36.9, +123.2]	0.993	—	1/11

Snapshot A has 3 post-training dates (no test is possible). Snapshot B was superseded by the longer V1 window below.

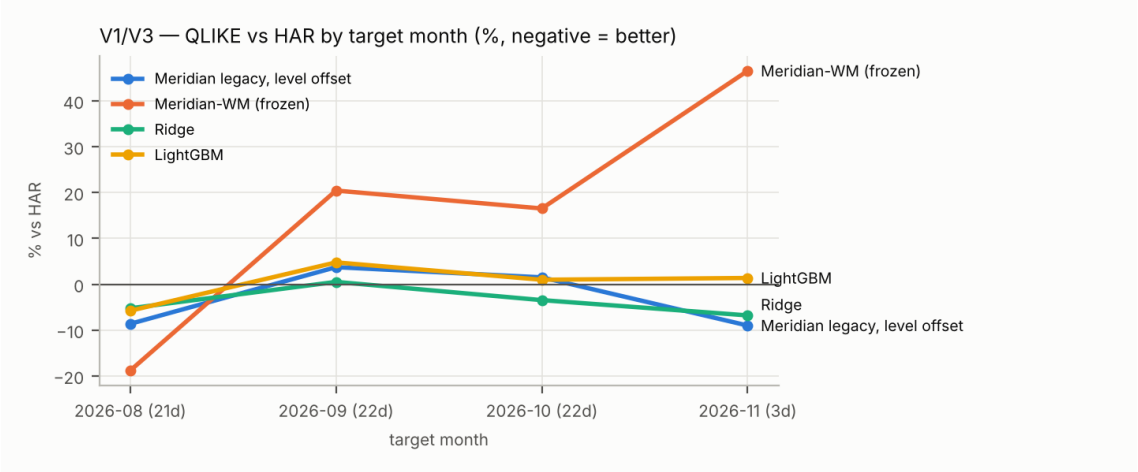
6. V1 + V3: frozen models over the longest post-training window

All models' training data end with labels on **2026-07-29** (legacy model: last origin reconstructed from its saved scaler, audit T4; frozen WM: 53997 training rows, labels through 2026-07-29). Scored targets: **2026-08-03** → **2026-11-04**, 68 dates × 11 assets = 737 forecasts. Nothing was retrained during the window.



Rank	Model	QLIKE	vs ref.	95% CI	DM p	MCS 90%	Assets beating ref.
1	Ridge	0.2994	-3.0%	[-6.8, +1.4]	0.093	in	9/11
2	Random forest	0.3011	-2.5%	[-5.1, +1.3]	0.039	in	10/11
3	Meridian legacy, level offset	0.3041	-1.5%	[-5.9, +4.7]	0.349	in	6/11
4	LHAR	0.3048	-1.3%	[-4.0, +1.2]	0.139	in	7/11
5	HAR-X (+VIX)	0.3071	-0.5%	[-4.7, +3.5]	0.408	in	7/11
6	HAR-RV pooled	0.3075	-0.4%	[-2.0, +1.3]	0.260	in	6/11
7	LightGBM	0.3087	-0.0%	[-6.0, +7.3]	0.450	in	7/11
8	HAR-RV per asset (ref.)	0.3088	+0.0%	—	—	in	0/11
9	SHAR	0.3098	+0.3%	[-1.3, +2.2]	0.616	in	6/11
10	EWMA	0.3171	+2.7%	[-5.1, +13.4]	0.728	in	5/11
11	TimeMixer	0.3247	+5.1%	[-1.8, +12.6]	0.929	out	2/11
12	Meridian-WM (frozen)	0.3332	+7.9%	[-8.0, +21.5]	0.877	in	3/11
13	22-day mean	0.3374	+9.3%	[-0.7, +23.6]	0.949	in	4/11
14	Meridian legacy, Jensen term	0.3607	+16.8%	[-3.7, +35.5]	0.954	in	2/11
15	GJR-GARCH	0.4095	+32.6%	[+18.4, +49.4]	1.000	out	1/11
16	GARCH(1,1)	0.4265	+38.1%	[+21.3, +57.7]	1.000	out	1/11
17	EGARCH	0.4300	+39.3%	[+21.4, +64.1]	1.000	out	2/11
18	Random walk	0.4664	+51.1%	[+35.9, +66.9]	1.000	out	0/11
19	Meridian legacy, raw head	0.5309	+71.9%	[+35.2, +104.1]	1.000	out	0/11

DM: one-sided Newey–West test that the model's date-level loss is below HAR's. CI: moving-block bootstrap over dates. MCS: Hansen model confidence set at 90%. Legacy level offset and Jensen term were estimated on pre-cut-off rows only (bias +0.393, Jensen 0.284 log units); the legacy model was trained on uncleaned pre-audit FX bars, which biases its raw head low.



Per asset — QLIKE vs per-asset HAR

Asset	Meridian legacy, level offset	Meridian-WM (frozen)	Ridge	LightGBM	GJR-GARCH
AAPL	+8.6%	+37.2%	-0.1%	-3.1%	+32.2%
EURUSD	-4.9%	-6.4%	-2.6%	-2.7%	-0.3%
GBPUSD	-7.5%	-14.7%	-3.9%	-8.1%	+194.0%
IWM	-8.0%	+5.0%	+2.3%	+4.6%	+24.9%
JNJ	-5.0%	+16.4%	+6.7%	+22.3%	+4.7%
JPM	-2.6%	+14.3%	-4.9%	-8.8%	+14.0%
MSFT	+5.0%	+23.0%	-0.1%	+0.2%	+49.9%
QQQ	+1.9%	+14.3%	-7.5%	-5.9%	+8.1%
SPY	-4.9%	-2.1%	-6.0%	-4.4%	+8.7%
USDJPY	+2.0%	+4.6%	-7.4%	+5.4%	+9.8%
XOM	+0.3%	+15.9%	-5.1%	-2.8%	+1.3%

Date alignment sensitivity (audit D4)

Yahoo labels FX bars one day early during UK summer time, so on label dates FX and equity rows of different trading days were averaged together (79 'dates' instead of 68 trading days). The tables above use the corrected trading-day alignment; the label-date version is shown for transparency. QLIKE values are identical; only date-level tests move.

Model	DM p (label dates)	DM p (trading days)	CI (label dates)	CI (trading days)
Meridian legacy, level offset	0.183	0.349	[-5.7, +4.9]	[-5.9, +4.7]
Meridian-WM (frozen)	0.676	0.877	[-7.2, +21.3]	[-8.0, +21.5]
Ridge	0.019	0.093	[-6.8, +1.6]	[-6.8, +1.4]
Random forest	0.006	0.039	[-5.1, +1.4]	[-5.1, +1.3]
LightGBM	0.212	0.450	[-6.1, +7.7]	[-6.0, +7.3]

7. V1 + V3: tail risk (next-day VaR / Expected Shortfall)

Volatility forecasts are turned into VaR/ES by filtering returns through them and reading the tail from past standardized returns (FHS; EVT for the legacy model, the Amendment-7 selection). Reference: GJR-GARCH + FHS. FZ0 is the joint VaR/ES scoring loss (lower is better). Kupiec / Christoffersen counts are assets passing at 5%.

2.5% level — 737 asset-days, 68 dates

Model	FZ0	Δ vs GJR	DM p	Violations (target 2.5%)	Pooled Kupiec p	Kupiec pass	Christoffersen pass
HAR-RV + FHS	-3.7920	-0.0473	0.137	3.26%	0.208	11/11	11/11
LightGBM + FHS	-3.7898	-0.0451	0.122	3.12%	0.298	11/11	11/11
Meridian-WM Student-t head	-3.7639	-0.0193	0.347	2.58%	0.893	11/11	11/11
Meridian-WM + FHS	-3.7565	-0.0119	0.341	2.58%	0.893	11/11	11/11
GJR-GARCH + FHS (ref.)	-3.7446	+0.0000	—	3.12%	0.298	11/11	11/11
Meridian legacy + FHS	-3.7371	+0.0075	0.535	2.99%	0.413	11/11	11/11
Meridian legacy + EVT	-3.7309	+0.0138	0.588	2.99%	0.413	11/11	11/11

1.0% level — 737 asset-days, 68 dates

Model	FZ0	Δ vs GJR	DM p	Violations (target 1.0%)	Pooled Kupiec p	Kupiec pass	Christoffersen pass
HAR-RV + FHS	-3.6085	-0.0726	0.120	0.95%	0.890	11/11	11/11
LightGBM + FHS	-3.6075	-0.0716	0.074	1.09%	0.818	11/11	11/11
Meridian legacy + FHS	-3.5775	-0.0416	0.203	1.09%	0.818	10/11	11/11
Meridian-WM + FHS	-3.5771	-0.0413	0.151	0.95%	0.890	11/11	11/11
Meridian legacy + EVT	-3.5684	-0.0325	0.222	0.95%	0.890	11/11	11/11
Meridian-WM Student-t head	-3.5673	-0.0314	0.308	0.81%	0.600	11/11	11/11
GJR-GARCH + FHS (ref.)	-3.5359	+0.0000	—	0.41%	0.066	11/11	11/11

With about 67 days per asset a single asset expects only ~1.7 violations at 2.5%, so per-asset coverage tests have very little power; the pooled Kupiec test and FZ0 differences are the informative columns. Tail DM tests use the trading-day alignment.

8. V2: Meridian-WM in the audited walk-forward harness

Primary result (as pre-registered): FAILED

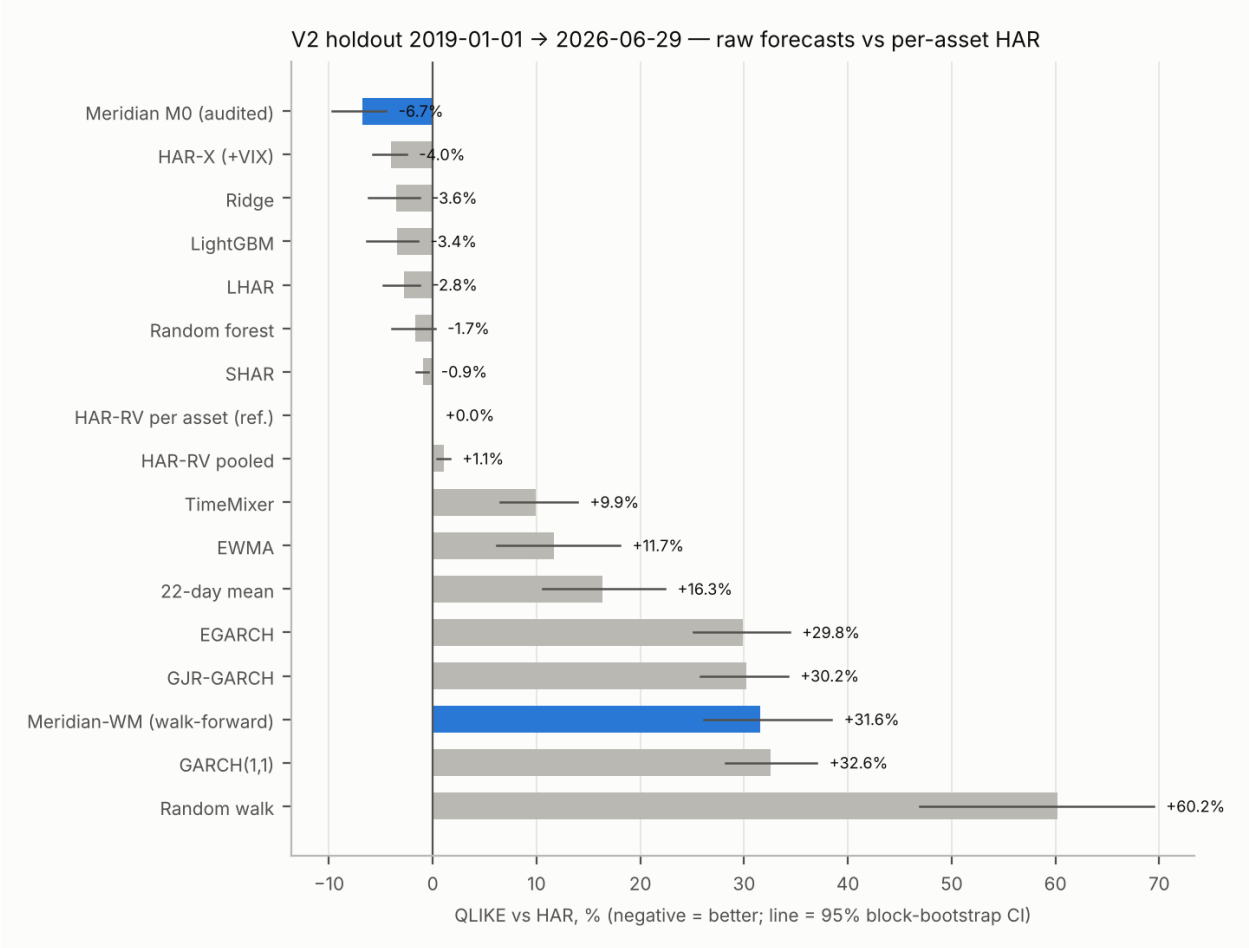
Meridian-WM exactly as written could not complete the walk-forward. On fold 7 (test from 2019-04-25) training diverged to NaN for 3 of 3 seeds, after 6 folds had completed. Diagnosis (`scripts/diag_wm_nan.py` reproduces it): scripts/diag_wm_nan.py, seed 700: all loss terms finite until optimizer step 131, where the SIGReg term is NaN (QLIKE 0.042, JEPA 0.187 finite); gradient norm NaN; after the step every switching-core parameter is NaN and all outputs are non-finite at step 132. Root cause: meridian/model.py sigreg_loss standardizes each random projection by its batch std (+1e-6); when the representation collapses along a projection the 4th powers overflow; clip_grad_norm_ cannot bound a NaN gradient.

Secondary result (Amendment 8a, committed before the re-run): same model and loss; optimizer steps whose loss or gradient norm is non-finite are skipped and counted. Healthy steps are untouched.

Skipped (non-finite) optimizer steps per fold

Seed	F1	F2	F3	F4	F5	F6	F7	F8	F9	F10	F11	F12	Total s kipped
seed0	0	0	0	0	0	0	8	0	0	0	0	0	8 of 11928
seed1	0	0	0	0	0	0	8	0	0	0	0	0	8 of 11928
seed2	0	0	0	0	0	0	8	0	0	0	0	0	8 of 11928

Meridian-WM (default configuration, 3 seeds, 8 epochs) was trained per fold on the same 12 expanding folds, 22-day embargo, label purge and train-only scaling as every model in the audited benchmark. Fold grid identity: 49421/49717 benchmark rows matched, 0 fold mismatches. On the same rows the rescored HAR and M0 holdout QLIKE equal the stored benchmark exactly (HAR 0.3249 → 0.3249; M0 0.3029 → 0.3029).



holdout: raw forecasts — 20898 forecasts, 2168 dates

Model	QLIKE	vs HAR-RV per asset (ref.)	95% CI	DM p	MCS 90%	IC	Assets beating ref.
Meridian M0 (audited)	0.3030	-6.7%	[-9.7, -4.5]	<0.001	in	0.592	11/11
HAR-X (+VIX)	0.3119	-4.0%	[-5.7, -2.5]	<0.001	out	0.579	10/11
Ridge	0.3133	-3.6%	[-6.2, -1.3]	<0.001	out	0.580	10/11
LightGBM	0.3138	-3.4%	[-6.3, -1.4]	0.003	out	0.583	10/11
LHAR	0.3159	-2.8%	[-4.8, -1.3]	0.001	out	0.570	10/11
Random forest	0.3195	-1.7%	[-4.0, +0.2]	0.039	out	0.573	8/11
SHAR	0.3218	-0.9%	[-1.6, -0.4]	0.008	out	0.566	7/11
HAR-RV per asset (ref.)	0.3249	+0.0%	—	—	out	0.562	0/11
HAR-RV pooled	0.3285	+1.1%	[+0.4, +1.6]	1.000	out	0.563	2/11
TimeMixer	0.3570	+9.9%	[+6.5, +13.9]	1.000	out	0.542	0/11
EWMA	0.3627	+11.7%	[+6.2, +18.0]	1.000	out	0.516	2/11
22-day mean	0.3779	+16.3%	[+10.6, +22.4]	1.000	out	0.502	0/11
EGARCH	0.4218	+29.8%	[+25.1, +34.4]	1.000	out	0.514	0/11
GJR-GARCH	0.4231	+30.2%	[+25.8, +34.2]	1.000	out	0.505	0/11
Meridian-WM (walk-forward)	0.4275	+31.6%	[+26.1, +38.4]	1.000	out	0.533	0/11
GARCH(1,1)	0.4307	+32.6%	[+28.2, +37.0]	1.000	out	0.489	0/11
Random walk	0.5205	+60.2%	[+46.9, +69.5]	1.000	out	0.475	0/11

holdout: raw, FX aligned to trading day — 20898 forecasts, 1952 dates

Model	QLIKE	vs HAR-RV per asset (ref.)	95% CI	DM p	MCS 90%	IC	Assets beating ref.
Meridian M0 (audited)	0.3030	-6.7%	[-9.6, -4.5]	<0.001	in	0.592	11/11
Ridge	0.3133	-3.6%	[-6.0, -1.3]	0.002	out	0.580	10/11
LightGBM	0.3138	-3.4%	[-6.0, -1.4]	0.006	out	0.583	10/11
HAR-RV per asset (ref.)	0.3249	+0.0%	—	—	out	0.562	0/11
Meridian-WM (walk-forward)	0.4275	+31.6%	[+25.9, +38.8]	1.000	out	0.533	0/11

holdout: walk-forward recalibrated — 20898 forecasts, 2168 dates

Model	QLIKE	vs HAR-RV, recalibrated (ref.)	95% CI	DM p	MCS 90%	IC	Assets beating ref.
Meridian M0 (audited)	0.3030	-8.0%	[-12.7, -4.6]	<0.001	in	0.592	11/11
Meridian M0, recalibrated	0.3045	-7.5%	[-11.3, -4.7]	<0.001	in	0.591	11/11
LightGBM, recalibrated	0.3118	-5.3%	[-9.3, -2.5]	<0.001	out	0.583	10/11
HAR-RV per asset (ref.)	0.3249	-1.3%	[-3.7, +0.2]	0.085	out	0.562	11/11
HAR-RV, recalibrated (ref.)	0.3292	+0.0%	—	—	out	0.562	0/11
Meridian-WM, recalibrated	0.3517	+6.8%	[+3.1, +11.2]	1.000	out	0.534	1/11
Meridian-WM (walk-forward)	0.4275	+29.9%	[+25.2, +35.1]	1.000	out	0.533	0/11

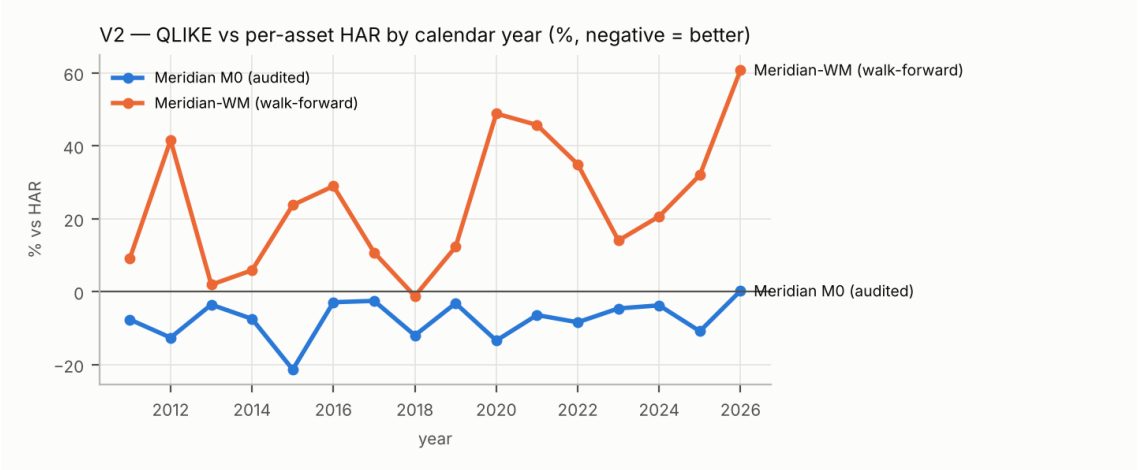
dev: raw forecasts — 20981 forecasts, 2183 dates

Model	QLIKE	vs HAR-RV per asset (ref.)	95% CI	DM p	MCS 90%	IC	Assets beating ref.
TimeMixer	1.2100	-33.2%	[-39.7, +10.6]	0.172	in	0.516	2/11
22-day mean	1.2136	-33.0%	[-41.4, +24.4]	0.187	in	0.493	2/11
EWMA	1.4296	-21.1%	[-27.7, +26.0]	0.204	in	0.504	2/11
LightGBM	1.4577	-19.6%	[-21.7, -3.1]	0.145	in	0.568	9/11
Random forest	1.4901	-17.8%	[-19.9, -1.7]	0.149	in	0.560	9/11

Model	QLIKE	vs HAR-RV per asset (ref.)	95% CI	DM p	MCS 90%	IC	Assets beating ref.
EGARCH	1.5566	-14.1%	[-22.2, +45.2]	0.276	in	0.489	1/11
Meridian M0 (audited)	1.5952	-12.0%	[-13.2, -4.1]	0.116	in	0.573	10/11
GJR-GARCH	1.6642	-8.2%	[-15.2, +44.5]	0.331	out	0.467	3/11
HAR-RV pooled	1.6653	-8.1%	[-9.5, +1.5]	0.165	in	0.545	3/11
GARCH(1,1)	1.6960	-6.4%	[-13.9, +47.6]	0.370	out	0.441	1/11
LHAR	1.7965	-0.9%	[-7.4, -0.1]	0.037	in	0.557	9/11
Ridge	1.7974	-0.8%	[-7.7, -0.0]	0.107	in	0.561	9/11
HAR-RV per asset (ref.)	1.8122	+0.0%	—	—	in	0.542	0/11
HAR-X (+VIX)	1.8220	+0.5%	[-6.9, +1.5]	0.614	in	0.559	9/11
SHAR	1.8911	+4.4%	[-2.1, +5.3]	0.826	in	0.549	8/11
Random walk	2.2387	+23.5%	[+17.9, +76.1]	0.991	out	0.453	0/11
Meridian-WM seed 0	2.3795	+31.3%	[+8.1, +34.4]	0.862	in	0.518	1/11
Meridian-WM seed 1	2.4018	+32.5%	[+18.9, +34.7]	0.888	in	0.493	0/11
Meridian-WM (walk-forward)	2.4725	+36.4%	[+9.5, +40.3]	0.860	in	0.533	0/11
Meridian-WM seed 2	2.7481	+51.6%	[+18.2, +56.1]	0.867	in	0.509	0/11
WM_scale	9.0600	+400.0%	[+90.6, +2711.0]	1.000	out	0.551	1/11

Loss concentration (a finding about the original benchmark's development period)

In the development period the 10 worst rows carry 82% of HAR's total QLIKE (20981 rows); in the holdout they carry 4%. Largest development-period rows: GBPUSD 2012-01-26 (next-day variance 0.4079, QLIKE 27003); EURUSD 2012-01-26 (next-day variance 0.1503, QLIKE 3454); JPM 2015-08-21 (next-day variance 0.0260, QLIKE 195). The 2012-01-26 FX bars imply a one-day variance far outside anything else in the series and look like a bad price tick that the bar cleaner (which only removes high <= low bars) does not catch. Development-period rankings, which the original project used to choose its configuration, are therefore driven by a handful of rows; holdout results are not. This is reported, not corrected, here.



Holdout by asset

Asset	Meridian-WM vs HAR	Meridian M0 vs HAR
AAPL	+42.1%	-7.3%
EURUSD	+16.6%	-4.2%
GBPUSD	+28.3%	-4.1%
IWM	+31.7%	-11.0%
JNJ	+42.4%	-3.7%
JPM	+42.2%	-9.0%
MSFT	+36.8%	-8.5%
QQQ	+25.9%	-12.6%
SPY	+32.0%	-10.3%
USDJPY	+24.5%	-1.1%
XOM	+37.4%	-8.0%

Seed dispersion (holdout, raw, vs HAR): Meridian-WM seed 0 +42.3%, Meridian-WM seed 1 +40.5%, Meridian-WM seed 2 +36.5%, WM_scale +2598.7%.

Meridian-WM Student-t head — next-day return VaR coverage on the holdout

Level	Violation rate	Asset-days	Kupiec pass	Christoffersen pass
2.5%	2.18%	20898	7/11	7/11
1.0%	0.90%	20898	7/11	7/11

Regime branch on the holdout: mean dwell 44.3 days; share by regime 0: 51%, 2: 35%, 1: 14%; mean next-day log RV by regime 0: -9.58, 1: -10.27, 2: -9.82. Descriptive only; no regime-existence claim is made.

9. Live pre-committed log and the audit

Forecast log: 104 rows, 65 pre-committed (logged before their outcome existed) over 6 origin dates, 2026-10-30 → 2026-11-10. Scored so far: 4 origins, Meridian M0 QLIKE 0.5213 vs frozen HAR 0.5042 (+3.4%). Rows for days when the logger did not run are kept but labelled not pre-committed and never scored. Under ~20 scored dates the result is noise.

ID	Check	Status	Key evidence
W1	weights_unchanged	PASS	{"results/forward/frozen/meridian_M0_seed0.pt": {"matches_manifest": true, "matches_git_HEAD": true}, "results/forward/frozen/meridian_M0_seed1.pt": {"matches_manifest": true, "matches_git_HEAD": true}, "results/forward/frozen/mer…
W2	no_training_code_path	PASS	{"banned_calls_found": [], "all_meridian_in_eval_mode": true}
T1	targets_after_training	PASS	{"A": {"train_label_end_max": "2026-10-30", "first_scored_target": "2026-11-02", "calendar_gap_days": 3, "rows_violating": 0}, "B": {"train_label_end_max": "2026-09-08", "first_scored_target": "2026-10-01", "calendar_gap_days": 23…
T2	baselines_same_cutoff	PASS	{"A": {"meridian_label_end": "2026-10-30", "baselines_label_end": "2026-10-30", "baselines_origin_max": "2026-10-29", "ok": true}, "B": {"meridian_label_end": "2026-09-08", "baselines_label_end": "2026-09-08", "baselines_origin_ma…
T3	lookahead_truncation	PASS	{"origins_tested": 24, "max_abs_diff_log_var": {"Meridian_M0": 0.0, "Meridian_legacy_raw": 9.5367431640625e-07, "HAR_frozen": 0.0, "EWMA_state": 0.0, "GARCH_state": 0.0}, "tolerance": 0.0001}
T4	legacy_provenance	PASS	{"best_match_training_data": "uncleaned (pre-audit) bars", "best_match_last_origin": "2026-07-28", "max_scaled_error": 0.00013085266239996364, "assumed_cutoff_used_for_baselines": "2026-09-08", "eval_window_starts": "2026-10-01", …
D1	data_source_parity	PASS	{"SPY": {"source": "yahoo", "overlap": 3794, "median_abs_log_diff": 0.0, "corr_log": 0.9999809964933933}, "QQQ": {"source": "yahoo", "overlap": 3794, "median_abs_log_diff": 0.0, "corr_log": 0.9999999843703453}, "IWM": {"source": "…
D2	truth_recomputed	PASS	{"rows": 361, "max_abs_log_diff": 0.0, "target_date_mismatches": 0}
D3	bars_complete	PASS	{"SPY": {"last_bar": "2026-11-04", "after_last_complete_day": false, "duplicates": 0, "degenerate_bars_in_window": 0}, "QQQ": {"last_bar": "2026-11-04", "after_last_complete_day": false, "duplicates": 0, "degenerate_bars_in_window…
D4	fx_bar_dating_no_leak	PASS	{"fx_bars_labelled_one_day_early": {"EURUSD": 3025, "USDJPY": 3024, "GBPUSD": 3040}, "V1_V3_last_train_label_true_day": "2026-07-30", "V1_V3_first_scored_fx_target_true_day": "2026-08-04", "A_first_scored_fx_target_true_day": "202…
R1	precommitted_reproduced	PASS	{"logged_rows_recomputed": 39, "max_abs_diff_Meridian_realtime_vintage": 9.5367431640625e-07, "max_abs_diff_HAR": 1.7763568394002505e-15, "max_abs_diff_deployed_same_day_macro": "0.21962643", "note": "deployed forecasts use the sa…
R2	frozen_har_reproduced	PASS	{"max_abs_diff_log_var": 1.0658141036401503e-14}
S1	scores_recomputed	PASS	{"numbers_checked": 72, "max_abs_diff": 0.0}
S2	report_matches_json	PASS	{"identical_to_fresh_render": {"DEPLOYMENT.md": true, "README.md": true}}
S3	no_tests_on_tiny_samples	PASS	{"violations": []}
S4	scorer_sanity	WARN	{"oracle_QLIKE": 0.0, "A:Meridian_M0": {"QLIKE": 0.2853426168283866, "shuffled_mean": 0.2525753566041285, "share_shuffles_better": 0.985}, "A:HAR": {"QLIKE": 0.31228182935757837, "shuffled_mean": 0.2787036217676103, "share_shuffle…
X1	preregistered_before_results	PASS	{"amendment8_committed": "2026-11-05 05:24:45+00:00", "results_created": {"v1v3_run": "2026-11-05T05:54:41+00:00", "frozen_wm": "2026-11-05T05:54:22+00:00", "wm_walkforward_combined": "2026-11-10T02:42:37+00:00"}}
X2	frozen_wm_unchanged	PASS	{"results/wm/frozen/wm_seed0.pt": {"matches_manifest": true, "matches_git_HEAD": true}, "results/wm/frozen/wm_seed1.pt": {"matches_manifest": true, "matches_git_HEAD": true}, "results/wm/frozen/wm_seed2.pt": {"matches_manifest": t…
X3	v1v3_targets_after_training	PASS	{"training_ends": {"legacy_B_last_origin(T4)": "2026-07-28", "baselines_label_end": "2026-07-29", "frozen_wm_label_end": "2026-07-29"}, "first_scored_target": "2026-08-03", "eval_rows_violating": 0, "eval_rows": 871}
X4	v1v3_lookahead_truncation	PASS	{"origins_tested": 12, "max_abs_diff_log_var": {"MeridianWM_frozen": 9.5367431640625e-07, "Meridian_legacy_raw": 9.5367431640625e-07}}
X5	v1v3_scores_recomputed	PASS	{"numbers_checked": 33, "max_abs_diff": 0.0}
X6	wm_walkforward_fold_grid	PASS	{"rows_bench": 49717, "rows_wm": 49421, "rows_matched": 49421, "fold_mismatch": 0, "only_in_bench": 296, "only_in_wm": 0, "nonfinite_forecasts": 0}
X7	wm_walkforward_rescored_and_reproduces_benchmark	PASS	{"rescore_max_abs_diff": 1.1102230246251565e-16, "stored_benchmark_max_abs_diff": 1.1631812180112888e-10, "same_rows_as_stored_benchmark": true, "rows": 20898}

Audit run 2026-11-11T01:50:52+00:00 on commit b522968. Full evidence: results/deploy/audit_report.json. The WARN (S4) is a finding, not a defect: within these short windows shuffling a model's forecasts across days scores as well as the true order, for HAR as well, so day-to-day timing skill cannot be shown yet.

10. Defects found and fixed during the work

• **Real-time VIX vintage.** Live forecasts used the previous day's VIX because FRED publishes with a delay; the deployment used same-day VIX. Both are reported; the logged forecasts are reproduced exactly from their vintage (R1).

• **Incomplete FX bars.** Yahoo FX bars are UTC days; a run before 00:00 UTC would log a partial bar. `forward_test.py` now drops it (unit test added).

• **Kupiec p-value index.** The new tail scorer first read the violation rate instead of the p-value from `kupiec()`; fixed before any number was reported.

• **Audit X5 crash.** A boolean-negation bug in the independent FZ0 recomputation; fixed, check now passes.

• **Over-claim in a summary.** An early README sentence said every interval contained zero; two did not. Corrected.

• **FX bar dating (pre-existing since the project's first data loader).** Yahoo stamps daily FX bars at London midnight; normalizing the UTC timestamp labels every summer-time bar one day early (Sunday–Thursday). It is not a look-ahead leak (each series stays in order and the VIX joined to those rows is older, not newer; audit D4 checks every train/test boundary on the true day), but date-level significance tests mixed FX and equity rows from different days, which overstated significance (Section 6), and the forward test would have accepted an open FX bar in summer. Fixed with meridian.data.fx_true_day; tests align by trading day.

• **Scheduled logger without repository access.** Fresh scheduled sessions could not push; the daily task now runs in the working session. Days it could not run are recorded as missed, never back-filled.

11. Limitations and what is not claimed

• Post-training windows are short (3 dates for the audited M0; 66 for the legacy model and frozen WM); intervals are wide.

• V2 used 3 seeds for Meridian-WM against 5 for the stored M0 (CPU budget). Seed dispersion is reported.

• Daily range-based realized variance only; no intraday data. Free, delayed data.

• A better volatility forecast is not by itself a profitable strategy; no trading or Sharpe number is claimed.

• The regime branch is descriptive; no claim that discovered regimes exist.

12. Reproduce

```
python scripts/deploy_data.py && python scripts/deploy_frozen.py && python scripts/score_deploy.py

python scripts/validate_v1v3.py run && python scripts/validate_v1v3.py score

for k in 0 1 2; do python scripts/benchmark_wm.py $k; done && python scripts/benchmark_wm.py combine && python scripts/score_wm.py

python scripts/audit_deploy.py && python -m pytest tests -q

python scripts/build_validation_pdf.py
```

Frozen file	sha256
results/wm/frozen/wm_seed0.pt	b8a244e80930be6f534182fe6e5d30bc1ebd5358adf0354f4d0627ad19562f5f
results/wm/frozen/wm_seed1.pt	7c01a8b9ab233f66ebc7b102e4006e6af61d4f3d2a44657729a121ca419c945d
results/wm/frozen/wm_seed2.pt	9df6acbdc460c8c4126375f5ca5595c0a5c0a480fb93af11a6871aa3b4464ebd
results/wm/frozen/wm_seed3.pt	3f3456fdc768f037e5bf6d39a3e8e43eb1adefc8e4704284c9118610243d733f
results/wm/frozen/wm_seed4.pt	d32d49353eeabf47b2e3fce97ccce19d5b055c7d3858f3c8badffd973aaf5303
results/wm/frozen/scaler.npz	c3374e67a3ddd9a1bc1cfc75cc79f061d842d555a95dccbafe430cc1e138fbe9
results/forward/frozen/meridian_M0_seed0.pt	d95d1b2914dd05f50798828581f112f805ea4d1984617a8c89975004240c69bf
results/forward/frozen/meridian_M0_seed1.pt	8ed958efc1e53b2e7d85347d56d5a83b4afbb0e6486dab47f1e0fc962a058dbd
results/forward/frozen/meridian_M0_seed2.pt	05613c5bab81b3d41ea2cac8081061580f3ae755e9bfbaa764cb706ccb91d33e
results/forward/frozen/meridian_M0_seed3.pt	ed90b6e92ba0052ed3b595d265ef6fcf7468f9a0f7c1ae57b30ae5cc8d5c376b
results/forward/frozen/meridian_M0_seed4.pt	57d976a33bb8a38ae9fc3b958281ab020d18fafd63b65d816211bad0f48c2daa
results/forward/frozen/scaler.npz	0859965b75b10aed2c3233c3acc59c84f3db72187efc79c43e2bc28c018b5bde
results/forward/frozen/har.json	f5133e57523a0652ddf31e3f1201737540cb00006bc6e7d76ff268829430f699
results/meridian_final.pt	608a21d102bfd8f28e5739459941eee275076e51601470681b301baa3b562829